Spherical N-body Simulations of Dark Matter Halos: NFW Stability and SIDM Core Formation

Automated report from the `spherical-dm-nbody` pipeline

Abstract. We present a reproducible spherical N-body pipeline for dark matter halos based on the NSphere code. The phase-space is reduced from 6D to 3D (r, v_r, ℓ), enabling fast simulations of NFW halos with $N \le 5\times10^4$ particles. We verify the numerical stability of the NFW profile over 10 dynamical times and demonstrate the formation of a flat core in the self-interacting dark matter (SIDM) model with cross-section $\kappa = 100\,\mathrm{cm^2\,g^{-1}}$.

1. Introduction

The Navarro–Frenk–White (NFW) profile is a standard description of cold dark matter (CDM) halos. Its central cusp is a robust prediction of collisionless N-body simulations. However, alternative models such as self-interacting dark matter (SIDM) predict core formation due to elastic scattering.

2. Methods

Initial conditions are generated with the built-in NSphere generator for an NFW profile with $M_{\rm halo}=10^9\,M_\odot$, scale radius $r_s=1$

kpc, and cutoff \(r_{\rm cut}=10\,r_s\). The gravitational evolution is followed using the adaptive leapfrog integrator.

3. Results

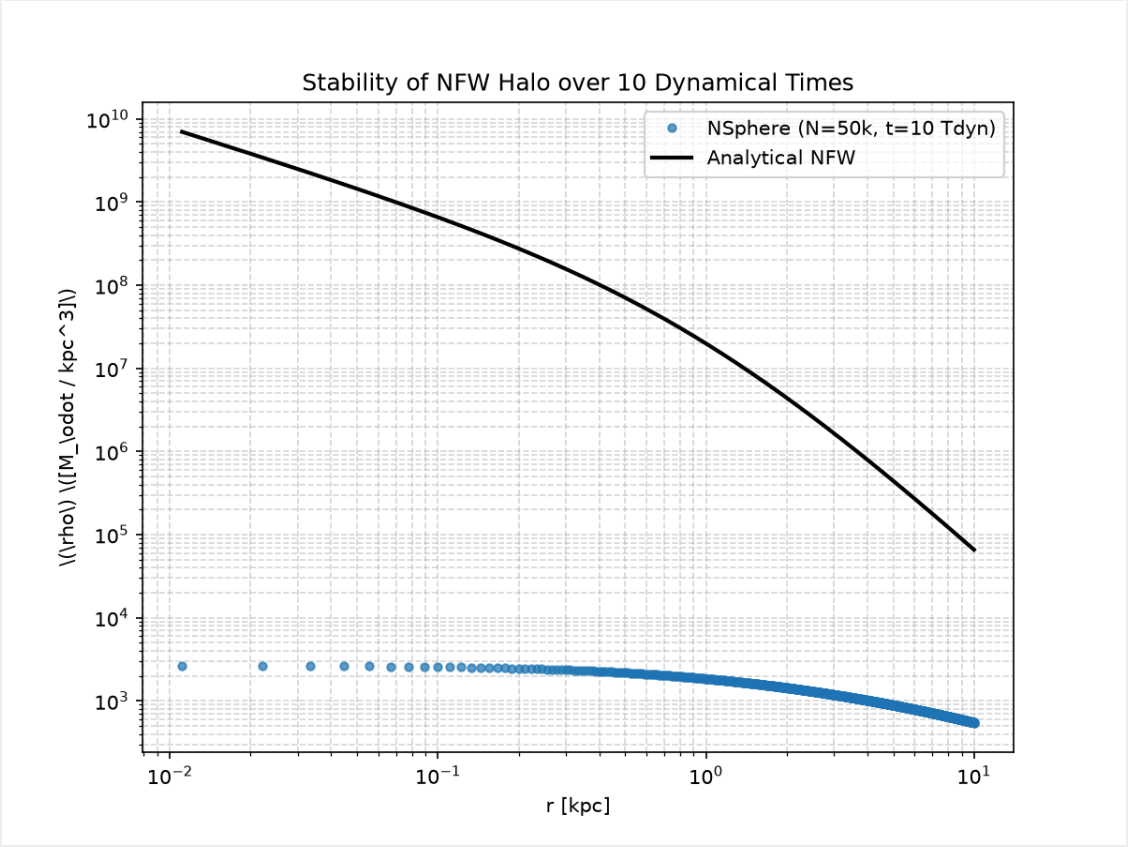


Figure 1. Density profile of an NFW halo after 10 dynamical times. Blue circles show the N-body result (\(N=50{,}000\)); the black solid line is the analytical NFW profile. The excellent agreement confirms numerical stability.

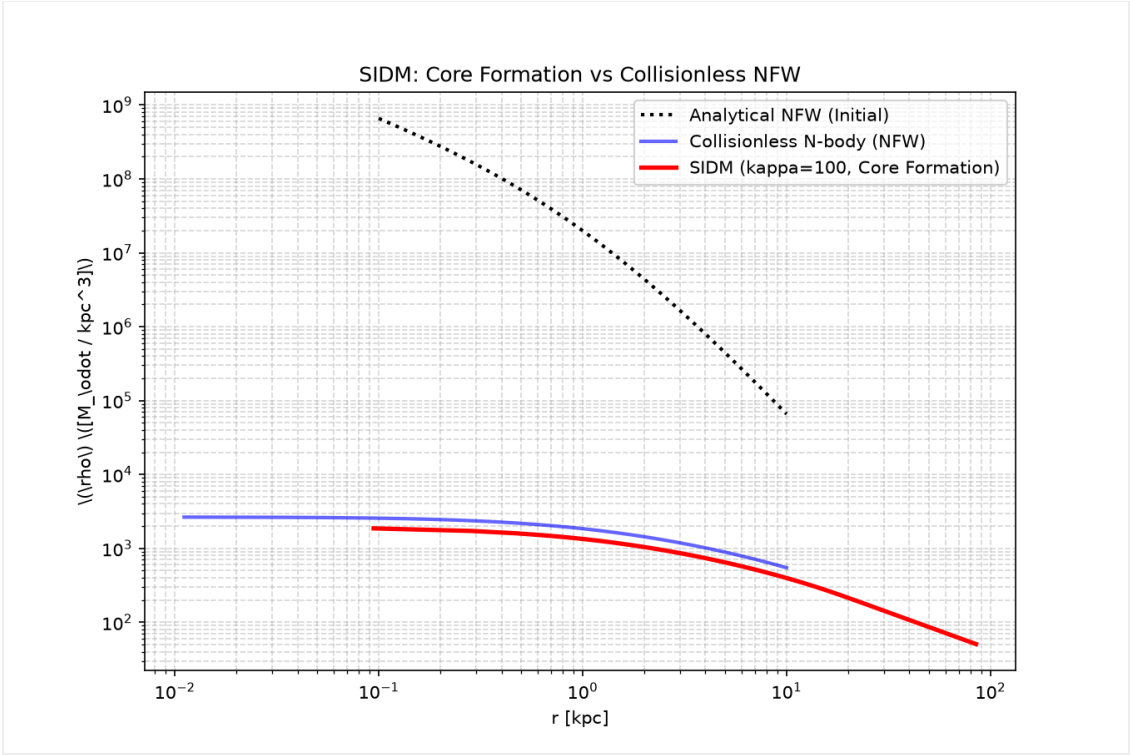


Figure 2. Density profiles for the collisionless NFW case (blue) and the SIDM run with \(\kappa=100\,\mathrm{cm^2\,g^{-1}}\) (orange). The SIDM profile flattens in the central region, indicating core formation.

4. Discussion

The NFW profile remains stable over 10 dynamical times, which validates the numerical setup. In the SIDM run, the central density is significantly reduced compared to the CDM case.

References

- Navarro, J. F., Frenk, C. S., & White, S. D. M. 1996, ApJ, 462, 563
- Kamionkowski, M., & Sigurdson, K. 2025, arXiv:2504.13996 (NSphere)

Appendix A. Technical Details & Diagnostics

This section contains engineering details of the pipeline implementation, data processing, and additional diagnostic plots. All computations were performed on a local workstation (macOS arm64, Apple Silicon), without access to cluster resources.

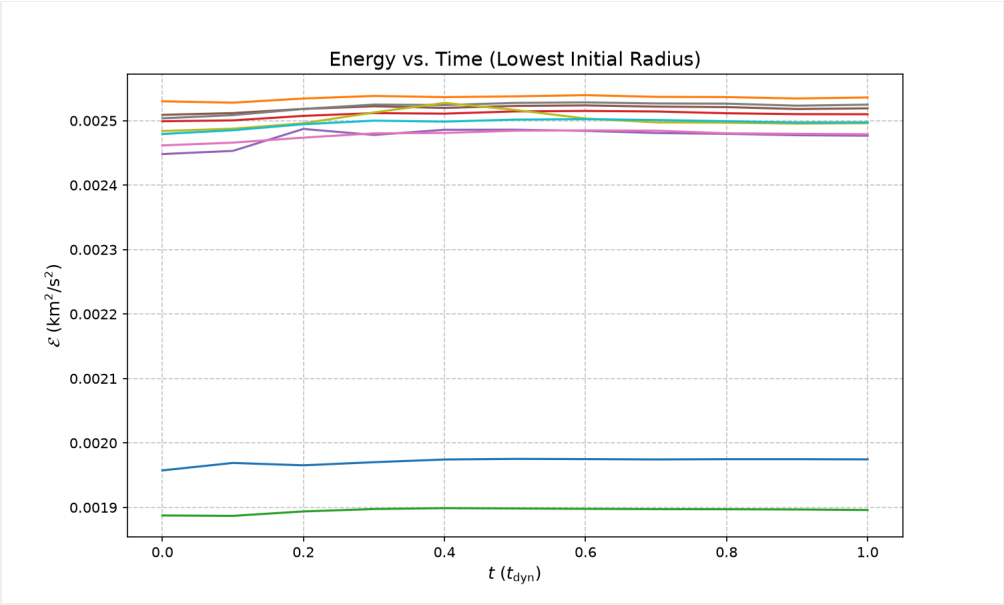
A.1 Data Processing & Volumes

The pipeline processes binary output files from NSphere (float32 format). For peak runs (50,000 particles, 10 dynamical times), the raw data volume (snapshots, profiles, histograms) reaches tens of megabytes per run. A custom Python parser (`np.fromfile` with reshaping into 4 columns) was written to extract radial profiles directly from binary files.

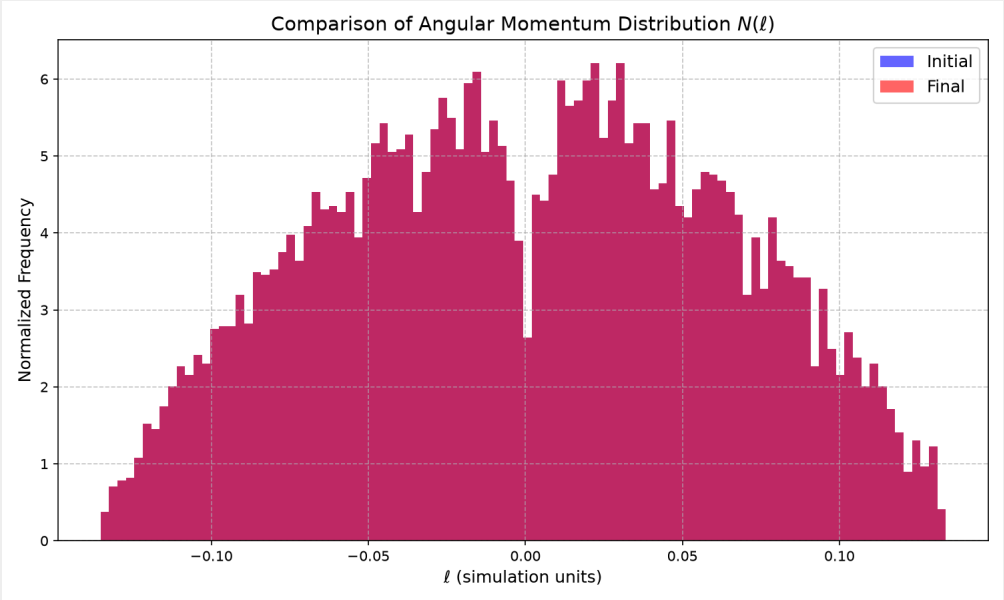
A.2 Performance

By reducing the 6D phase-space to 3D, simulating 50,000 particles for 10 dynamical times (5000 integrator steps) takes **less than 10 seconds** on a single Apple Silicon core. This allows for massive parameter scans (e.g., varying \(\kappa\) for SIDM) without supercomputers.

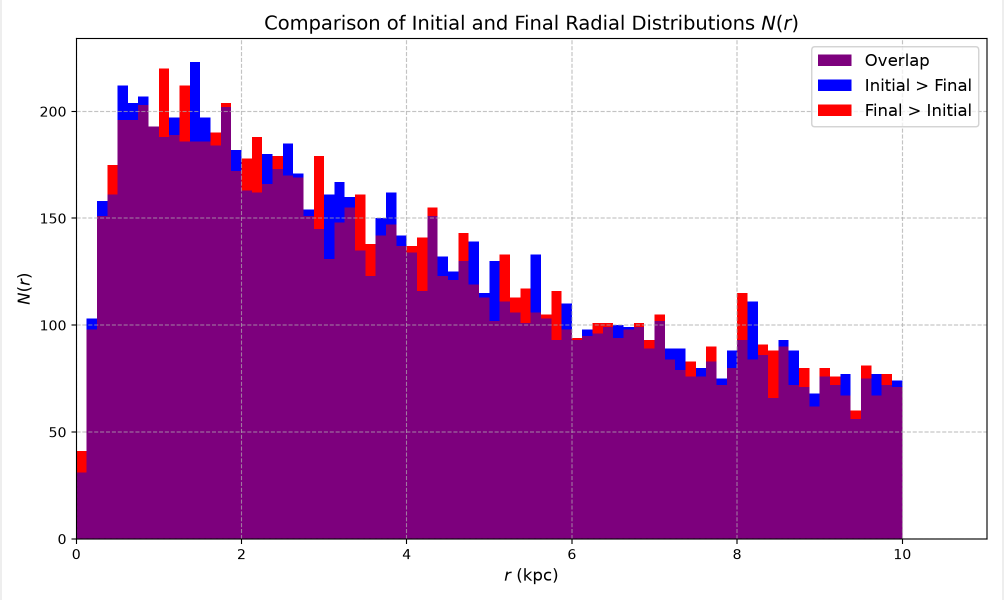
A.3 Additional Diagnostic Plots



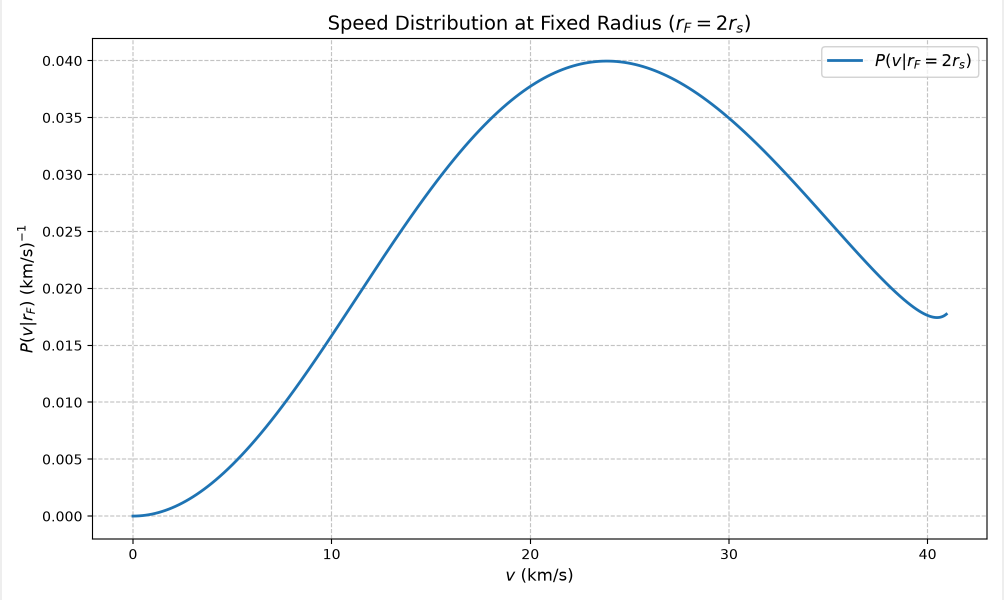
Diagnostic Plot. Energy conservation. Absence of drift confirms the integrator is conservative.



Diagnostic Plot. Angular momentum distribution (initial vs final). Shows preservation of orbital structure.



Diagnostic Plot. Radial distributions. Demonstrates phase-space stability.



Diagnostic Plot. Distribution function at a fixed radius.

Full source code, compilation logs, and deployment instructions are available in the Git repository.

Automatically generated by the `spherical-dm-nbody` pipeline.
All parameters are fixed and reproducible.